Anas Almasri

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Education

University of Toronto | 3.90 CGPA | Dean's List Scholar (2025, 2026) **Expected Graduation: 04/28**
Bachelor of Science in Computer Science and Mathematical Sciences (Candidate) *Toronto, ON*

Relevant Courses: Machine Learning, Neural Networks & Deep Learning, Artificial Intelligence, Data Structures & Algorithms, Computational Complexity, Probability & Statistics, Linear Algebra II, Advanced Calculus, Complex Analysis

Publications

M. Attarian, I. Vyse, J. Gerigk, E. Opryshko, Y. Ruan, **A. Almasri**, S. Singh, Y. Du, I. Gilitschenski, C. Voelcker. *Update-Free On-Policy Steering via Verifiers*. arXiv:2603.10282, 2026.

Research Experience

Toronto Intelligent Systems Lab (TISL), University of Toronto **Jun 2025 – Jan 2026**
Student Researcher – Robot Learning *Toronto, ON*

- **Co-author** of *Update-Free On-Policy Steering via Verifiers*, co-authored with **Google DeepMind** researchers: test-time steering of frozen diffusion/flow policies using verifiers trained on the policy's own rollouts.
- Method improved success rate by **49% on average across 5 real ALOHA tasks** with no policy fine-tuning.
- Built the **sim-to-real data pipeline**: rollout collection in **MuJoCo/Robomimic** and multi-camera preprocessing with OpenCV and NumPy.
- Wrote a custom **PyTorch** DataLoader that cut preprocessing time by **50%** across 10+ datasets.
- Adapted **Diffusion Policy** training code to custom simulation data; ran **ablation studies** isolating base-policy failure modes that informed verifier design.

Research Opportunity Program (MAT299) – Real Two-Dimensional Rational Maps **May 2026 – Present**
Supervisor: Prof. Nadya Askaripour, University of Toronto *Toronto, ON*

- Co-authoring a **journal in preparation** on the dynamics of a two-parameter planar rational map, with Prof. Askaripour and two student coauthors.
- Wrote rigorous proofs (orbits, global stability) and validated them with Python simulations and phase portraits.

Industry Experience

MongoDB **Jun 2026 – Aug 2026**
Software Engineering Intern – Docs Platform *Toronto, ON*

- Designed and shipped an in-house **LLM-powered translation service** processing **60,000+ files**, replacing a legacy vendor and cutting **$200k+ in annual costs**.
- Built an **LLM-as-a-judge evaluation suite** to score translation quality at scale, enabling continuous regression testing across languages.
- Instrumented the pipeline with **Braintrust** to track accuracy, latency, and failure rates in production.

Amicare **Jan 2026 – Apr 2026**
Software Engineering Intern *Toronto, ON*

- Built a full-stack healthcare platform (**React, Node.js, FastAPI**) serving **60+ franchise clinics**.
- Shipped patient intake and scheduling features that halved manual admin overhead for clinic staff.

Projects

AIAI – Autonomous Multi-Agent ML System | *PyTorch, Sklearn, FastAPI, Redis, LangChain* **2026**

- Designed a multi-agent system that converts natural-language task descriptions into end-to-end training pipelines via a finite-state-machine orchestrator.
- Achieved **80% average model accuracy** across vision and NLP tasks and **cut manual ML setup time by 90%+**.

Transformer Visualizer | *FastAPI, TransformerLens, NumPy, Next.js, React* | demo | code **2026**

- **Led a 5-person GDSC team** building an interactive explorer of **GPT-2 and BLOOM-560M** internals.
- TransformerLens backend serves real per-layer activations, per-head attention, and next-token probabilities across 12 layers, plus hallucination-detection metrics.

Technical Skills & Research Interests

Languages: Python, C++, C, Java, JavaScript, TypeScript, Bash
ML & Robotics: PyTorch, TensorFlow, TransformerLens, NumPy, OpenCV, MuJoCo, Robomimic, ROS 2, LangChain
Tools: Linux, Git, MongoDB, FastAPI, Docker, DigitalOcean, React, Next.js, Firebase, Supabase, Netlify, Vercel
Research Interests: robot learning, inference-time steering, mechanistic interpretability, LLM evaluation
Interests & Hobbies: Vexillology (Flags), Educating, Sports, Travelling (14 countries)